Acute Sore Throat and Fusobacterium Necrophorum in PHC – A Systematic Review and Meta-analysis - Appendix 1 and 2
Version: (11 January 2021)

Appendix 1: Search strings for PubMed and PROSPERO

PubMed:

(("pharyngitis"[MH] OR "pharyngitis"[TIAB] OR "Lemierre Syndrome"[MH] OR "Lemierre Syndrome"[TIAB] OR "Necrobacillosis"[TIAB] OR "pharyngotonsillitis "[TIAB] OR "tonsillitits"[TIAB] OR "throat"[TIAB] OR "epidemiology"[TIAB])
AND ("fusobacterium necrophorum"[MH] OR "fusobacterium"[TIAB]))
NOT "Case Reports"[pt]

Scopus:

(TITLE-ABS-KEY ((pharyngitis OR "Lemierre Syndrome" OR necrobacillosis OR pharyngotonsillitis OR tonsillitis OR "Peritonsillar Abscess" OR throat OR epidemiology))
AND TITLE-ABS-KEY (fusobacterium))
AND DOCTYPE (ar)
AND (LIMIT-TO (SUBJAREA , "MEDI ") OR LIMIT-TO (SUBJAREA , " IMMU ")
OR LIMIT-TO (SUBJAREA , " BIO "))
AND (LIMIT-TO (LANGUAGE , "English "))
AND (EXCLUDE (EXACTKEYWORD , "Case Report"))

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Appendix 2: Formulae for calculating P-EPV

Formula for the point estimate of P-EPV:

$$P(\mathrm{D}^+\left|\mathrm{S}^+\mathrm{T}^+\right.) = 1 - \left(\frac{\frac{\mathrm{Sen}}{P(\mathrm{T}^+\left|\mathrm{S}^+\right.)} - 1}{\frac{\mathrm{Sen}}{P(\mathrm{T}^+\left|\mathrm{S}^+\mathrm{D}^-\right.)} - 1} \right)$$

Formula for the confidence interval of P-EPV:

$$\left(1 - \left(\frac{\frac{\mathrm{Sen}}{a} - 1}{\frac{\mathrm{Sen}}{d} - 1} \right) \right), \left(1 - \left(\frac{\frac{\mathrm{Sen}}{b} - 1}{\frac{\mathrm{Sen}}{c} - 1} \right) \right)$$

$$a = p(\mathrm{T}^+\left|\mathrm{S}^+\right.) - \left(Z_{\alpha/4} \times \sqrt{\frac{p(\mathrm{T}^+\left|\mathrm{S}^+\right.) \times \left(1 - p(\mathrm{T}^+\left|\mathrm{S}^+\right.)\right)}{\#(\mathrm{S}^+)}} \right)$$

$$b = p(\mathrm{T}^+\left|\mathrm{S}^+\right.) + \left(Z_{\alpha/4} \times \sqrt{\frac{p(\mathrm{T}^+\left|\mathrm{S}^+\right.) \times \left(1 - p(\mathrm{T}^+\left|\mathrm{S}^+\right.)\right)}{\#(\mathrm{S}^+)}} \right)$$

$$c = p(\mathrm{T}^+\left|\mathrm{S}^+\mathrm{D}^-\right.) - \left(Z_{\alpha/4} \times \theta \times \sqrt{\frac{p(\mathrm{T}^+\left|\mathrm{S}^-\right.) \times \left(1 - p(\mathrm{T}^+\left|\mathrm{S}^-\right.)\right)}{\#(\mathrm{S}^-)}} \right)$$

$$d = p(\mathrm{T}^+\left|\mathrm{S}^+\mathrm{D}^-\right.) + \left(Z_{\alpha/4} \times \theta \times \sqrt{\frac{p(\mathrm{T}^+\left|\mathrm{S}^-\right.) \times \left(1 - p(\mathrm{T}^+\left|\mathrm{S}^-\right.)\right)}{\#(\mathrm{S}^-)}} \right)$$

Definitions:
T = Test outcome which may be positive (T^+) or negative (T^-)
S = Symptoms of a sore throat which may be present (S^+) or absent (S^-)
M = The bacteria of interest (FN or GAS) which may be present (M^+) or absent (M^-)
D = Disease (a sore throat caused by FN/GAS) which may be present (D^+) or absent (D^-)
Sensitivity of test to detect bacteria = $\mathrm{Sen} = P(\mathrm{T}^+\left|\mathrm{M}^+\right.)$

Setting the width of the confidence interval. For 95%: $Z_{\alpha/4} = Z_{0.05/4} = Z_{0.0125} = 2.24$

An online calculator doing the above: https://science-network.tv/epv-calculator/

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